Thesis Plan

Performance, Safety, and Engineering Trade-offs of Rust and C++ in Nokia RAN Control-Plane Systems

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1 Background and motivation

Modern RAN control-plane software (the L3 of a gNB, e.g. the gNB-CU-CP [1]) is written mostly in C++, which delivers the throughput and latency such systems need but carries well-known memory-safety and concurrency risks (use-after-free, data races, double-free). These classes dominate real defect populations, and migrating to memory-safe languages is now an explicit policy position [2, 3]. Rust guarantees safety at compile time on machine-checked foundations [4].

Its fit for latency-critical telecom control planes is unproven. The "Rust performs like C++" consensus rests on general-purpose benchmarks [5] that report average-case behaviour. Closer studies find real costs, both in networking code [6] and in specific mechanisms such as bounds and overflow checking [7]. For a RAN control plane the operative metric is not the average but the tail [8]. This thesis is carried out at Nokia on a compact model of the per-UE state work a gNB-CU-CP does.

The decision has a long horizon. 3GPP has allocated Release 20 to the 6G studies and Release 21 to the normative work, with full specifications due to the ITU by mid-2030 and migration from 5G explicitly in scope [9, 10], so a language chosen for the control plane of the current generation is inherited by the next.

2 Research gap

The closest prior work is a cluster of at least five master's theses carried out at Ericsson. Their conclusions conflict. Karlbäck and Orö compare Rust and Go against C++ for RAN applications and conclude C++ remains the choice where performance is critical [11]. Olsson reaches the same domain from the management side and reports Rust's execution speed on par with or better than C's, but its peak memory higher [12]. Olsson and Stenström build a thread-per-core packet processor on asynchronous Rust and measure between 20% and 50% of the throughput of the DPDK application it mirrors [13]. Arvidsson and Kaller measure median and 99th-percentile latency for 5G control-plane procedures across five asynchronous frameworks: C++ with Boost.Fibers is fastest at every load. The Rust runtimes track the shape of its median curve, but their p99 rises more steeply past a threshold rate [14]. Emberg and Wadsten come closest in method, building matched C++ and Rust ASN.1 decompilers, reaching third-party C++ from Rust through a C wrapper, and comparing the two statistically [15].

Three of the five vary the runtime and concurrency model together with the language: Boost.Fibers, goroutines, Tokio, Monoio, Glommio, DPDK Eventdev. A measured difference therefore cannot be attributed to the language rather than to the framework built on it, and the effect sizes involved, up to a factor of five, indicate how much of the measurement the runtime owns. Independent real-time analysis of those runtimes supports that reading [16]. The

remaining two report execution time, and in one case peak memory, rather than tail behaviour under rising load.

Stated as requirements rather than as an absence, the gap is narrow, and each requirement is met somewhere. No single study meets them all. Matched implementations built from one design exist in systems software [6, 17, 18, 19] and, in telecom, in Emberg and Wadsten. Percentile latency for control-plane procedures is reported by Arvidsson and Kaller, and adoption effort is measured by Karlbäck and Orö. No study combines them for a stateful, concurrent control-plane subsystem: holding the specification, algorithms, concurrency model, allocator, and build settings fixed, *and* sweeping past the saturation knee, *and* measuring what incremental migration costs. All five theses are also unindexed grey literature, absent from the citation databases and reachable only through repository search, so the evidence a vendor would weigh this decision on has not been through review.

Two further properties of that evidence matter. It is concentrated in one vendor: the five studies use that vendor's frameworks and hardware targets, and none of their conclusions has been tested against another organisation's software, so their transferability is untested rather than established. And the record on the other side is emptier still. A search of Nokia's indexed publications returns no study of Rust, and none on memory safety in the radio access network. The nearest peer-reviewed position from that side, co-authored by two Nokia researchers, argues for making memory-error sanitisation compilers practical on the premise that certifying compilation is not [20], which is the after-the-fact detection strategy this thesis sets a compile-time guarantee against. The question is therefore open in the literature and unexamined in this setting, and the artefacts published with the thesis let an independent organisation check the answer.

3 Objective and research questions

The thesis builds two functionally equivalent implementations of a UE Context Management service in modern C++ and in Rust and compares them under a spec-justified synthetic signalling workload. Since similar average throughput is already known, it targets four less-explored dimensions:

RQ1 How do Rust and C++ compare in latency predictability (median, p99, p99.9, p99.99, maximum) under increasing load and overload, with the concurrency model held fixed?

RQ2 Which classes of C++ memory and concurrency defects does safe Rust eliminate at compile time, at what stage are the rest detected, and which classes does it *not* prevent?

RQ3 What overhead, if any, does Rust's safety model impose on throughput, CPU, and memory?

RQ4 What are the practical implications (effort, FFI interoperability, risk) of adopting Rust incrementally in an existing C++ RAN codebase?

RQ2 is deliberately two-sided: a catalogue containing only defects Rust prevents would confirm its own premise, so it spans the boundary: seven classes safe Rust rejects and seven it does not. Memory-safety defects in real Rust code invariably trace back to `unsafe` [21], which one boundary class now reproduces directly, while deadlock remains an open problem for the language [22]. Two of the seven boundary classes returned the opposite of the expected result, with the C++ tooling reporting what Rust's does not, which is the kind of finding a one-sided catalogue could not have produced.

4 Methodology

A single language-agnostic specification (public API, UE state machine, data structures, sharding and concurrency model) is frozen before any implementation code is written, so the comparison measures language rather than design differences. Both implementations use identical

algorithms, a shared-nothing sharded worker-thread model, the same allocator, and comparable build optimisation. Holding the concurrency model fixed matters because thread-per-core structuring itself changes tail latency [23]. Both must pass the same parity suite, whose expected results come from an independent third implementation of the specification rather than from either language, so that agreement between the two cannot be mistaken for correctness of both.

The workload is swept from nominal load to overload on pinned cores, with interleaved repeats and interval estimates rather than single figures [24]: medians with nonparametric bootstrap confidence intervals [25], and exact paired permutation tests corrected across the family by the Holm procedure [26]. That family is fixed before the campaign runs. A load picked after reading a sweep cannot enter it, and the clock diagnostics below sit outside it too. The reported campaign now runs on a dedicated bare-metal host with the frequency governor fixed. The earlier virtualised host is retained for comparison, because it could not resolve the far tail.

Latency is captured in high-dynamic-range histograms against an intended send schedule, so coordinated omission does not hide the tail below saturation [27, 28]. Past saturation that correction is part of what the clock times, because the producer writes its extra samples inside the timed path. A same-host control keeps the intended send time and the cap, and writes the fill after the timed window. The processor, the pin and `nohz_full` stay the same. That control removes the large high-load gaps and leaves a smaller Rust throughput lead. CPU per completed operation falls with the fill, so the large ratio is the recorder. The smaller lead tracks extra kernel time per handoff, which a queue swap on the reported campaign host isolates from the core call. A no-fill clock from an earlier host is coordinated-omission biased by construction and was not repeated here. The deferred-fill clock reports cumulative schedule drift rather than service time. Overload service latency is therefore left unresolved.

Evaluation adds a controlled fault-injection study [29] and an FFI prototype. On the FFI, the plan takes a narrow position: confining `unsafe` to the seam is a statement about what the compiler checks, not run-time protection, since host and component share an address space [30]. Only synthetic data is used, so no personal or confidential data is processed.

5 Expected contributions

- A reproducible, telecom-relevant benchmarking framework.
- A frozen shared specification and two parity-tested implementations.
- A quantitative comparison of latency and resource behaviour that divides at the saturation knee. Below it the far tail is resolved on bare metal. Past it extra delay samples are written after the timed window, so the study reports completed work and ranks neither language on service latency.
- A bound on what a comparison of this shape can attribute to a language at all, obtained by decomposing an end-to-end operation into the work the language performs and the cross-thread handoff around it, and by auditing what else the producer's timed path contains at high load. This one was not anticipated and emerged from the measurements.
- A defect-class taxonomy mapping both what the compile-time guarantee covers and where it stops.
- Evidence-based guidelines for incremental Rust adoption. The per-call cost at the FFI seam is measured and bounded on a real signalling call. The design and learning costs are named by prior work and are not measured here [31].

6 Status and timeline

Work is well under way. The shared specification, both implementations, the conformance suite, the fault-injection catalogue, the FFI prototype, and the benchmark harness are in place. Chapter drafts and a supervisor-read-through skeleton (`Thesis_v4.tex`) exist. The reported campaign now runs on a dedicated bare-metal host at the main sweep's own shard count, load

grid, repeat count and toolchain, and it resolves the RQ1 tail claim below saturation. A second bare-metal host at a different design agrees on every direction. The earlier virtualised campaign is retained for the host comparison. The reported host then repeated the grid with the histogram fill after the timed window. The processor, the pin and `nohz_full` were held. The large high-load gaps move with the fill. A smaller Rust lead remains on both heaps, and the visit does not explain it. The overload reading is narrowed to match. What remains is supervisor review and submission.

Item	State
Literature review	Drafted in thesis skeleton
Spec + implementations	Complete, frozen at v1.0 (`spec-v1.0`)
RQ2 (fault injection)	Complete
RQ4 (FFI prototype)	Complete
RQ1 / RQ3 (benchmarks)	Measured on one dedicated bare-metal EPYC host. The earlier virtualised campaign is retained for comparison, and a second bare-metal host at a different design confirms the directions. A same-host deferred-fill control isolates the large high-load gaps as recorder work
Thesis document	In progress (full draft + supervisor-facing skeleton)

Month (2026)	Goal	What happens
June	Foundations	Done: topic defined, C++/Rust/5G foundations, literature review drafted.
July	Draft	Done: thesis plan approved, chapter skeleton and evaluation design outlined.
August	Build and measure	Done: specification and both implementations built, parity suite and benchmark harness operational, reported bare-metal campaign run, earlier virtualised campaign retained for the host comparison, second bare-metal host at a different design confirming the directions.
September	Resolve	Same-host fill control on the reported bare-metal host, abstract, full proofread, draft to supervisors and Nokia instructors.
October	Review	Revise on supervisor and instructor feedback, prepare the colloquium.
November	Submit	Final revisions, colloquium presentation, submission.

Risks. The far tail is now resolved on bare metal below saturation. The same-host fill control is done: the large high-load gaps move with the recorder, and the smaller Rust throughput lead is tied to kernel time on the handoff path. One open question still needs no new machine. It is the transparent-hugepage candidate for the allocator effect, a two-setting comparison on one host with everything else pinned. The language effect on peak resident set size keeps its sign and its MiB scale on all three hosts measured. The size of the allocator effect does not travel between them. Single-author skill asymmetry is bounded by the frozen specification and parity suite, and acknowledged as a remaining performance limitation.

Out of scope. The full gNB, PHY/L2 and the user plane, real captured traces, managed-language alternatives, and production operational concerns.

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